Review notes · for you

Give instructions: unit check

Programming · Level 1 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Exact instructions
- Loops
- If this, then that
- Find and fix bugs

Activity: The robot relay

1. Warm-up (3 min): Say one idea you remember from each of the four lessons.
2. Plan (7 min): In teams, write a robot program to reach a flag on a chalk grid, with at least one loop.
3. Run (7 min): A pupil from another team is the robot and follows your program exactly.
4. Fix (3 min): If the robot misses, find the first wrong card and fix only that one.

Discussion

- Which lesson did you like best, and why?
- Which question was hardest?
- Where will you use loops or rules at home?

Check yourself

- One thing I can now tell a robot to do

Try it this week

Program a person at home: write robot cards to get them from the door to a chair. Did your program work the first time? If not, find the bug!

Quiz sheet

Give instructions: unit check

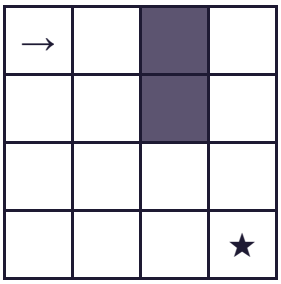
Programming · Level 1 · Unit check

Name _____ Class _____ Date _____

1 **What is an algorithm?**

◯ A kind of robot ◯ Exact steps, in order ◯ A fast computer

2 **Walk the robot around the wall to the flag, with 5 cards at most.**



Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

_____

_____

3 **"Forward × 4" moves the robot 4 squares.**

◯ True ◯ False

4 **Put the steps for planting a seed in order.**

☐ Cover it with soil

☐ Make a hole in the soil

☐ Water it

☐ Put the seed in the hole

5 **The lamp goes on IF button A OR button B is pressed. Nobody presses anything. Is the lamp on?**

◯ No ◯ Yes

6 **Match each long program to its loop.**

clap, clap, clap	• • • •	turn left × 4
forward, forward	• • • •	clap × 3
turn left, turn left, turn left, turn left	• • • •	forward × 2

7 **The robot must turn left, go forward 3 squares, turn right, and go forward 2. Put the program in order. One line doesn't belong.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

- [ ] `forward(2)`
- [ ] `left()`
- [ ] `forward(3)`
- [ ] `forward(5)`
- [ ] `right()`

8 **Which help you find a bug?**

- [ ] Throw all the cards away
- [ ] Find the first wrong card
- [ ] Run the program and watch
- [ ] Change one card at a time

9 **The robot should go forward 3, turn right, then go forward 2. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  forward(3)
2  left()
3  forward(2)
```

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10 **Which program is the shortest, but walks the same way?**

- ◯ forward, forward, forward, forward
- ◯ forward × 2, forward × 2
- ◯ forward × 4

Check your answers

Give instructions: unit check

Programming · Level 1 · Unit check

1 **Exact steps, in order**

An algorithm is a list of exact steps, in order, like a recipe or the robot's cards.

2 **Any program that takes the robot to the ★ with 5 cards or fewer.**

Turn right, forward 3 times, turn left, forward 3 times: four cards.

3 **True**

One card repeated 4 times moves 4 squares.

4 **Top to bottom: 3, 1, 4, 2**

Each step needs the one before it: you can't cover a seed you haven't put in.

5 **No**

OR needs at least one button. With none pressed, the lamp stays off.

6 **clap, clap, clap — clap × 3**
forward, forward — forward × 2
turn left, turn left, turn left, turn left — turn left × 4

Count how many times the step repeats, and write it once with that number.

7 **Top to bottom: 4, 1, 2, ✗, 3**

```
left()
forward(3)
right()
forward(2)
```

One line for each step of the walk, in the same order.

8 **Run the program and watch; Find the first wrong card; Change one card at a time**

Watch, find the first wrong card, and change one thing at a time.

9 **Line 2: `right()`**

Line 2 turns left, but the robot should turn right.

10 **forward × 4**

All three walk four squares. One card repeated 4 times is the shortest: a loop.